Preprints.org (www.preprints.org) | NOT PEER-REVIEWED | Posted: 6 October 2025 doi:10.20944/preprints202509.2421.v2

5.1. Preliminaries: Canonical Rank-One Objects and Normalization

Recall $\Sigma_\epsilon \equiv \Pi_{\rm PT}[(\partial\epsilon)^2]$, $\hat{n}_\mu \equiv \dfrac{\partial_\mu \epsilon}{\sqrt{|\Sigma_\epsilon|}}$, and the dimensionless, traceless rank-one matrix

$$\mathcal{T}^\mu{}_\nu \equiv \mathrm{sgn}(\Sigma_\epsilon)\Big(\hat{n}^\mu \hat{n}_\nu - \tfrac{1}{4}\delta^\mu{}_\nu\Big), \qquad \mathrm{Tr}\mathcal{T} = 0, \quad \mathrm{Tr}(\mathcal{T}^2) = \tfrac{3}{4}, \tag{31}$$

together with the dimension-one spurion scale (after C1),

$$\tau \equiv \hat{n}^\mu T_\mu \;=\; 3\,\eta\;\mathrm{sgn}(\Sigma_\epsilon)\,\sqrt{|\Sigma_\epsilon|}. \tag{32}$$

Key relation.

$$\tau^2 = 9\eta^2|\Sigma_\epsilon| = \frac{3}{2}\,(-\,\mathrm{sgn}(\Sigma_\epsilon) I_T), \qquad \mathrm{sgn}(\Sigma_\epsilon)\; I_T \equiv \mathrm{sgn}(\Sigma_\epsilon) I_T = -6\eta^2|\Sigma_\epsilon|. \tag{33}$$

5.2. Three One-Line Propositions (NY Split $\to *$ & Projection $\to$ Coefficient Match)

Proposition 7 (NY split). *$T^A \wedge T_A = \mathrm{NY} + e^A \wedge e^B \wedge R_{AB}$ with $\mathrm{NY} \equiv d(e^A \wedge T_A)$.*

Proposition 8 (*PT*-even shadow after $*$ and projection). *Under A2, the combined PT preserves orientation and satisfies $[\mathcal{PT}, *] = 0$. Applying $*$ to Proposition 7 and projecting to scalars, the term $*\mathrm{NY}$ contributes only as a boundary convention (A5), while the remaining PT-even bulk piece reduces, at quadratic order and* after C1, *to*

$$\delta^2\mathcal{L}_{\mathrm{CS}^+} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T + \nabla_\mu J^\mu_{\mathrm{CS}^+} \quad \textit{for some } A_\star \in \mathbb{R}.$$

Proposition 9 (Coefficient match under the σ_ϵ scheme). *With $\mathrm{Tr}\mathcal{T} = 0$, $\mathrm{Tr}(\mathcal{T}^2) = \frac{3}{4}$, and $\tau = 3\eta\;\mathrm{sgn}(\Sigma_\epsilon)\sqrt{|\Sigma_\epsilon|}$, both (i) the rank-one determinant route $\sqrt{\det[\mathbf{1} + (\frac{2}{3})\lambda\,\tau\,\mathcal{T}]}$ and (ii) the closed-metric deformation $\tilde{g}_{\mu\nu} = g_{\mu\nu} + (\frac{2}{3})\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu$ yield the same quadratic bulk coefficient $\delta^2\mathcal{L} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T$ with $A_\star = \lambda^2/8$.*

Proof (paper-level). Expand $\sqrt{\det(\mathbb{1} + X)} = 1 + \frac{1}{2}\mathrm{Tr}X + \frac{1}{8}[(\mathrm{Tr}X)^2 - 2\mathrm{Tr}X^2] + \mathcal{O}(X^3)$; with $X = (\frac{2}{3})\lambda\tau\mathcal{T}$, $\mathrm{Tr}\mathcal{T} = 0$ gives $\delta^2\sqrt{\det(\mathbb{1} + X)} = -\frac{1}{4}\mathrm{Tr}(X^2) = -\frac{1}{4}(\frac{4}{9})\lambda^2\tau^2\mathrm{Tr}(\mathcal{T}^2) = -\lambda^2\tau^2/12$. Using (33), $\delta^2\mathcal{L} = \sqrt{-g}(\lambda^2/8)\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T$. The closed-metric route has the same Jacobian $\sqrt{-\tilde{g}} = \sqrt{-g}\sqrt{\det[\mathbb{1} + (\frac{2}{3})\lambda\tau\mathcal{T}]}$, hence the identical coefficient.

5.3. Quick Derivations (ROD/CM/CS$^+$)

Rank-one determinant (rank-one determinant route).

$$\mathcal{L}_{\mathrm{ROD}} = \sqrt{-g}\Big(\sqrt{\det\big[\mathbf{1} + \tfrac{2}{3}\lambda\,\tau\,\mathcal{T}\big]} - 1\Big) \implies \boxed{\delta^2\mathcal{L}_{\mathrm{ROD}} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T\;,\; A_\star = \lambda^2/8}. \tag{34}$$

Closed-metric route. $\tilde{g}_{\mu\nu} = g_{\mu\nu} + \frac{2}{3}\lambda\,\tau\,\hat{n}_\mu\hat{n}_\nu \Rightarrow \sqrt{-\tilde{g}} = \sqrt{-g}\sqrt{\det[\mathbf{1} + \frac{2}{3}\lambda\,\tau\,\mathcal{T}]}$, so

$$\boxed{\delta^2\mathcal{L}_{\mathrm{CM}} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T\;,\; A_\star = \lambda^2/8}. \tag{35}$$

PT-even CS/Nieh–Yan shadow. By Proposition 8,

$$\boxed{\delta^2\mathcal{L}_{\mathrm{CS}^+} = A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T \;+\; \nabla_\mu J^\mu_{\mathrm{CS}^+},\;\; A_\star = \lambda^2/8}. \tag{36}$$

Bulk identity (summary).

$$\boxed{\delta^2\mathcal{L}_{\mathrm{ROD}} \;\equiv\; \delta^2\mathcal{L}_{\mathrm{CM}} \;\equiv\; \delta^2\mathcal{L}_{\mathrm{CS}^+} \;\equiv\; A_\star\,\sqrt{-g}\;\mathrm{sgn}(\Sigma_\epsilon)\; I_T \quad (\mathrm{mod}\;\nabla_\mu J^\mu),\; A_\star = \lambda^2/8}. \tag{37}$$

© 2025 by the author(s). Distributed under a Creative Commons CC BY license.